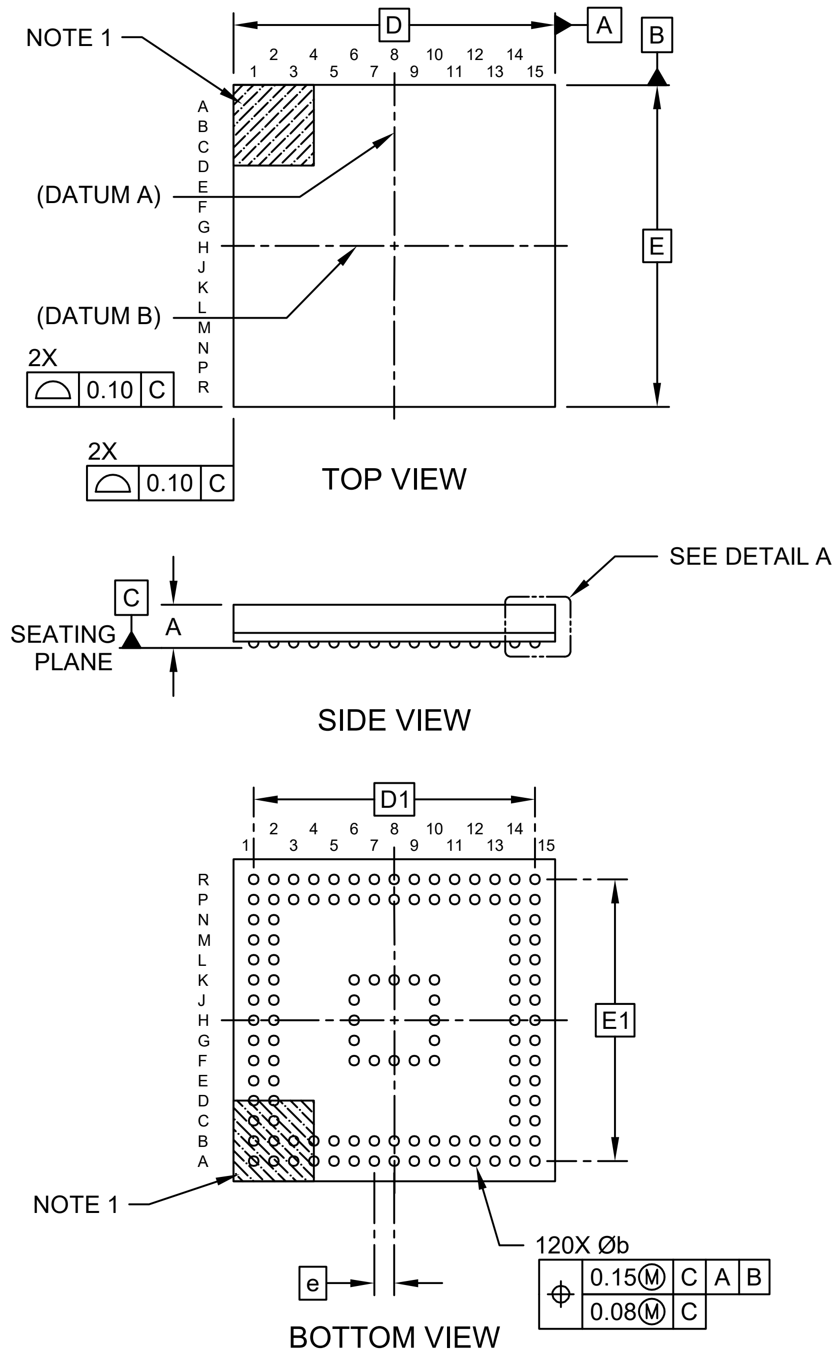
120-Ball Thin Fine Pitch Ball Grid Array Package (DGB) - 8x8 mm Body [TFBGA]

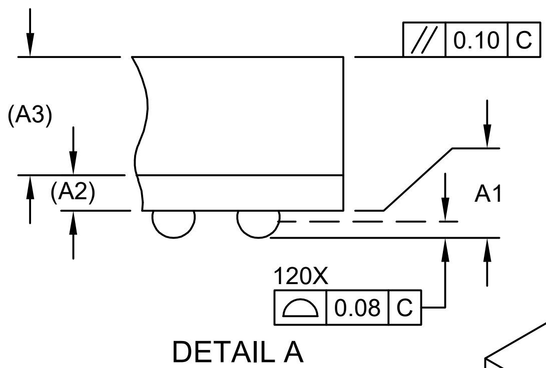
Note:	For the most current package drawings, please see the Microchip Packaging Specification located at http://www.microchip.com/packaging

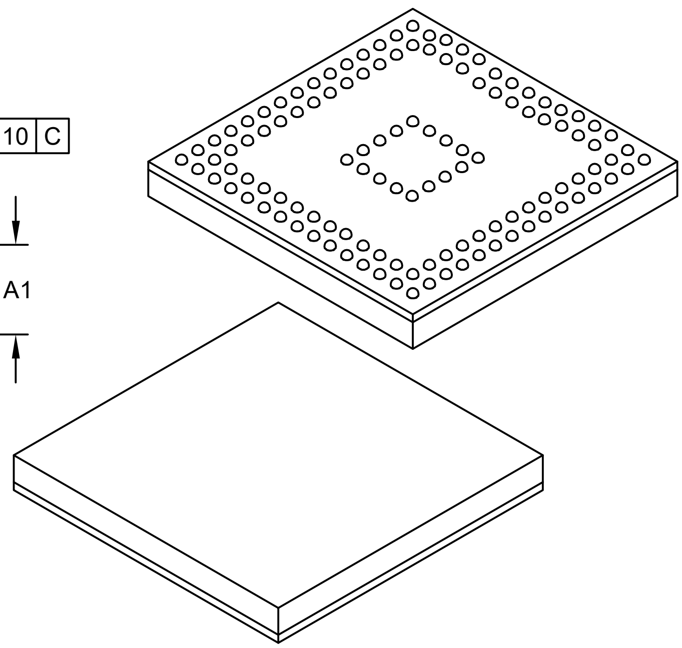


Microchip Technology Drawing C04-21465-DGB Rev C Sheet 1 of 2

120-Ball Thin Fine Pitch Ball Grid Array Package (DGB) - 8x8 mm Body [TFBGA]

Note:	For the most current package drawings, please see the Microchip Packaging Specification located at http://www.microchip.com/packaging





	Units	MILLIMETERS		
	Dimension Limits	MIN	NOM	MAX
Number of Terminals	N		120	
Pitch	e		0.50 BSC	
Overall Height	A	-	-	1.20
Standoff	A1	0.11	-	0.21
Substrate Thickness	A2		0.21 REF	
Mold Cap Thickness	A3		0.70 REF	
Overall Length	D		8.00 BSC	
Overall Ball Pitch	D1		7.00 BSC	
Overall Width	E		8.00 BSC	
Exposed Pad Width	E1		7.00 BSC	
Terminal Width	b	0.20	-	0.30

Notes:

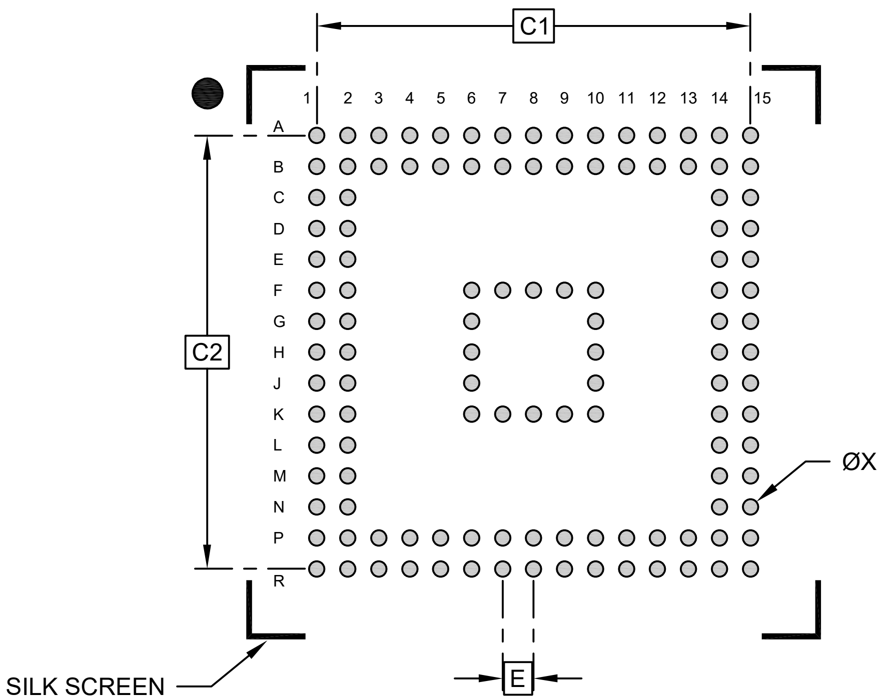
1. Pin 1 visual index feature may vary, but must be located within the hatched area.
2. Package is saw singulated
3. Dimensioning and tolerancing per ASME Y14.5M

BSC: Basic Dimension. Theoretically exact value shown without tolerances.
REF: Reference Dimension, usually without tolerance, for information purposes only.

Microchip Technology Drawing C04-21465-DGB Rev C Sheet 1 of 2

120-Ball Thin Fine Pitch Ball Grid Array Package (DGB) - 8x8 mm Body [TFBGA]

Note:	For the most current package drawings, please see the Microchip Packaging Specification located at http://www.microchip.com/packaging



RECOMMENDED LAND PATTERN

Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E		0.50 BSC	
Contact Pad Spacing	C1		7.00 BSC	
Contact Pad Spacing	C2		7.00 BSC	
Contact Pad Width (X20)	X		0.25	

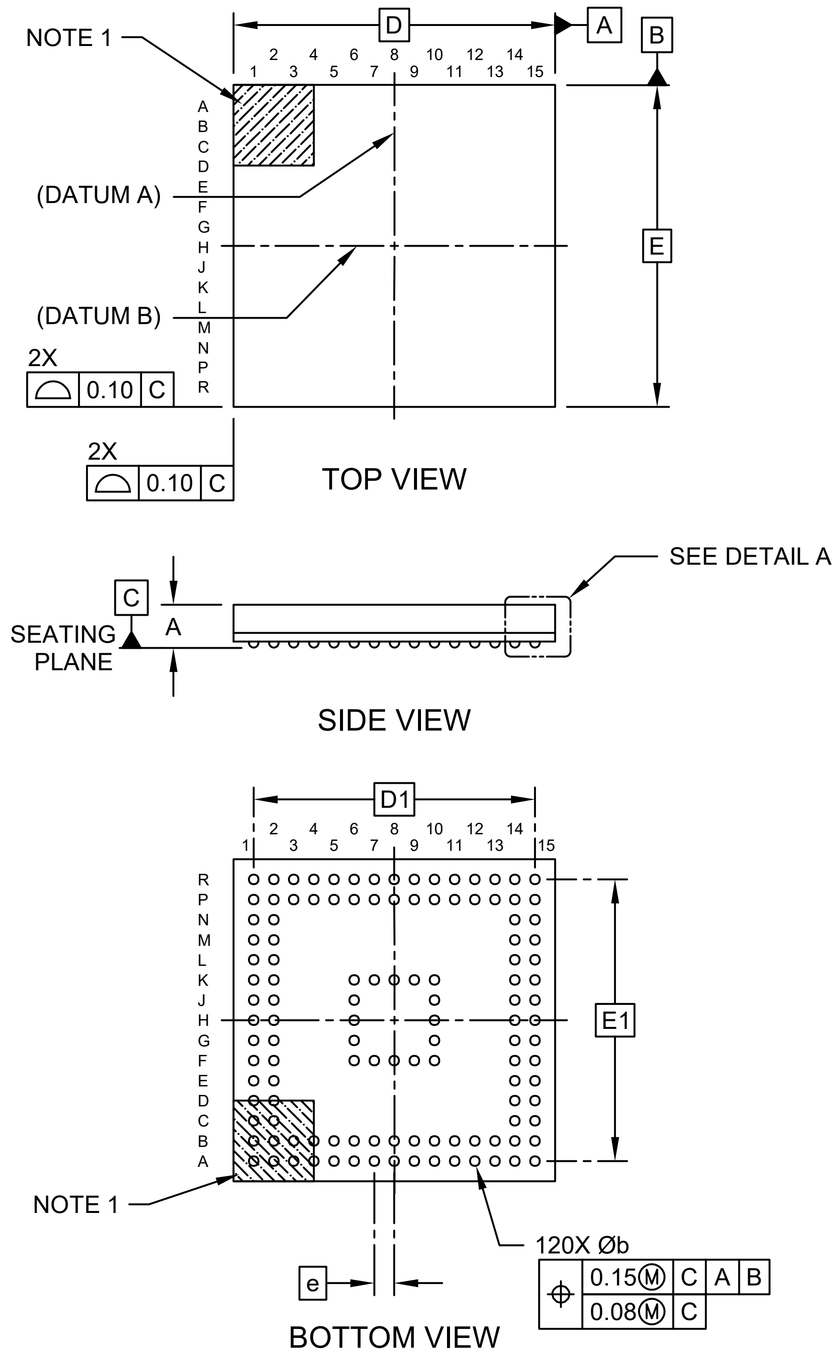
Notes:

1. Dimensioning and tolerancing per ASME Y14.5M
   BSC: Basic Dimension. Theoretically exact value shown without tolerances.

Microchip Technology Drawing C04-23465-DGB Rev C

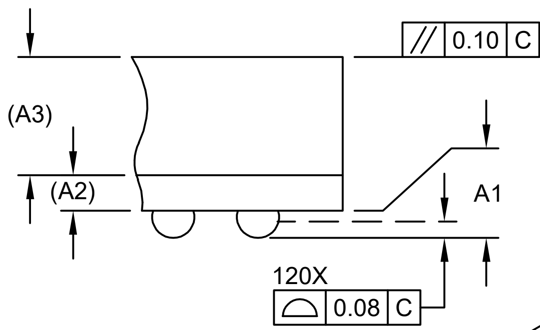
120-Ball Thin Fine Pitch Ball Grid Array Package (HEB) - 8x8 mm Body [TFBGA]

Note:	For the most current package drawings, please see the Microchip Packaging Specification located at http://www.microchip.com/packaging

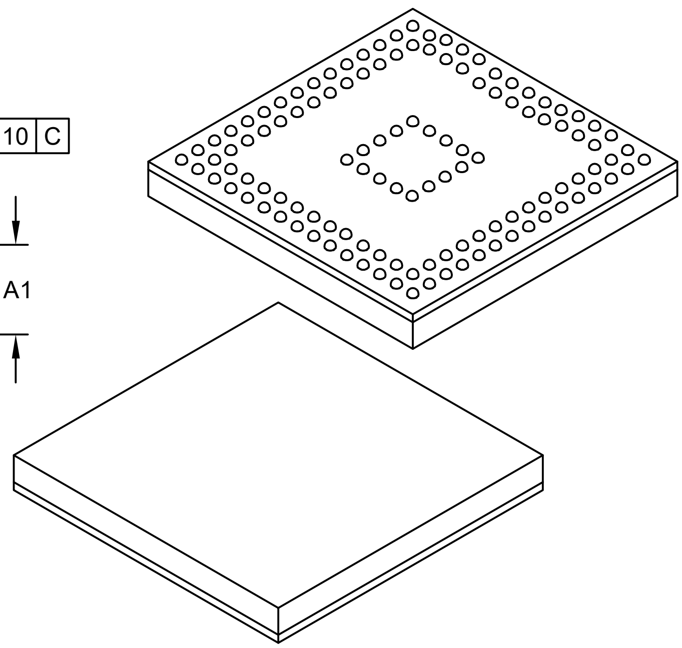


120-Ball Thin Fine Pitch Ball Grid Array Package (HEB) - 8x8 mm Body [TFBGA]

Note:	For the most current package drawings, please see the Microchip Packaging Specification located at http://www.microchip.com/packaging



DETAIL A



Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Number of Terminals	N		120	
Pitch	e		0.50 BSC	
Overall Height	A	-	-	1.20
Standoff	A1	0.11	-	0.21
Substrate Thickness	A2		0.21 REF	
Mold Cap Thickness	A3		0.70 REF	
Overall Length	D		8.00 BSC	
Overall Ball Pitch	D1		7.00 BSC	
Overall Width	E		8.00 BSC	
Exposed Pad Width	E1		7.00 BSC	
Terminal Width	b	0.20	-	0.30

Notes:

1. Pin 1 visual index feature may vary, but must be located within the hatched area.
2. Package is saw singulated
3. Dimensioning and tolerancing per ASME Y14.5M

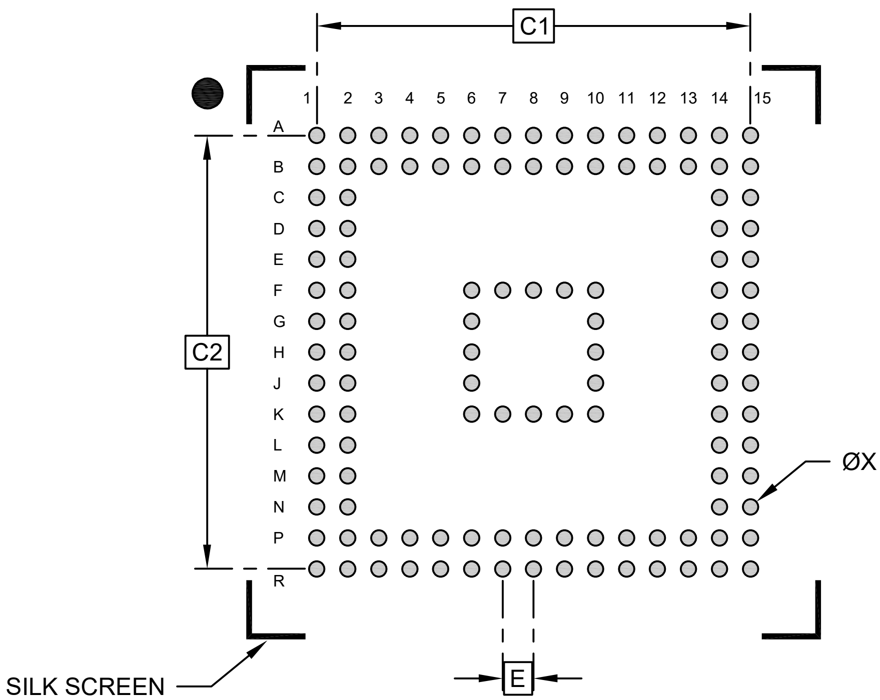
   BSC: Basic Dimension. Theoretically exact value shown without tolerances.

   REF: Reference Dimension, usually without tolerance, for information purposes only.

Microchip Technology Drawing C04-21465-HEB Rev C Sheet 1 of 2

120-Ball Thin Fine Pitch Ball Grid Array Package (HEB) - 8x8 mm Body [TFBGA]

Note:	For the most current package drawings, please see the Microchip Packaging Specification located at http://www.microchip.com/packaging



RECOMMENDED LAND PATTERN

Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E		0.50 BSC	
Contact Pad Spacing	C1		7.00 BSC	
Contact Pad Spacing	C2		7.00 BSC	
Contact Pad Width (X20)	X		0.25	

Notes:

1. Dimensioning and tolerancing per ASME Y14.5M
   BSC: Basic Dimension. Theoretically exact value shown without tolerances.

Microchip Technology Drawing C04-23465-HEB Rev C